

Data Collection and Preprocessing Phase

Date	1 August 2026
Team ID	Individual Project
Project Title	Food Ordering Behaviour Analysis
Maximum Marks	10 Marks

Data Exploration and Preprocessing Template

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

Section	Description
Data Overview	The dataset contains 50,000 food ordering records and 19 fields covering customer, order, location, cuisine, meal, restaurant, spending, delivery fee, rating, repeat-order, mood, hunger, company and rainy-weather information.
Data Cleaning	Checked missing values and duplicates. All 19 columns have 0 missing values and there are 0 duplicate rows. Numeric fields were checked for valid ranges, and categorical values were reviewed for consistency.
Data Transformation	Used filtering and sorting for city, cuisine and meal comparisons. Created an Age Group calculated field for analysis. Aggregated Order ID as order count and Order Value as total/average value. Reviewed order volume, revenue contribution and average delivery/order time by category.
Data Type Conversion	Kept order_id and user_id as identifiers, age/order_value/delivery_fee/time_taken_to_order/rating_given as numeric measures, and converted categorical text fields such as city, cuisine, meal_type and restaurant_type to dimensions/categories for Tableau analysis.
Column Splitting and Merging	No column splitting or merging is required because the source already stores the main business dimensions separately. The order_time field can be used directly as a categorical time-of-day dimension.
Data Modeling	The project uses one flat dataset, so no table relationships are required. Tableau calculated fields include Age Group, record count, and filter calculations for cuisine, city and meal type.
Save Processed Data	After validation, the cleaned dataset can be saved as a processed CSV for future Tableau use. No rows require deletion or imputation in the supplied dataset; the validated structure is retained.